

Clustering Assignment : Unsupervised Image Clustering

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Introduction

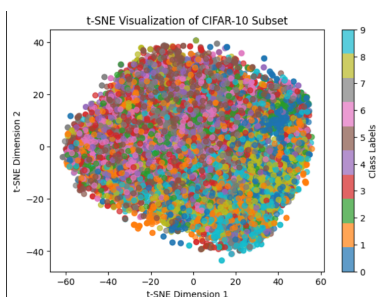
This assignment is centered on the application of clustering algorithms to the CIFAR-10 dataset based on both visual features from images and textual features obtained from captions. The overall goal is to compare and analyze the performance of clustering by utilizing visual features, textual features, and a combination of both modalities.

Dataset Description

The CIFAR-10 dataset was utilized, consisting of 60,000 color images distributed across 10 distinct classes: Airplanes(class 0) , Automobiles(class 1), Birds(class 2), Cats(class 3), Deer(class 4), Dogs(class 5), Frogs(class 6), Horses(class 7), Ships(class 8), Trucks(class 9).

There are 6,000 images in each class, and each image is of 32×32 pixel resolution in RGB. The dataset is very well balanced and has been used very commonly as a standard benchmark for image classification problems.

Images of CIFAR-10 Dataset



This **t-SNE visualization** represents a subset of CIFAR-10 images in a 2D space, with each point corresponding to an image and color-coded by its class label. The overlapping clusters indicate that different classes have similar visual features, making classification challenging. The spread suggests some separation, but the high-dimensional nature of image data leads to some mixing of classes.

For the assignment, the dataset was accessed using TensorFlow's own datasets library, here the given 10000 sample indices are chosen for clustering, allowing efficient loading and preprocessing. Although CIFAR-10 contains ground truth labels, they were only used for the purpose of evaluation. The clustering process itself was kept fully unsupervised.

Pre-processing Details

Visual Feature Extraction

In this approach, a **pre-trained MobileNetV2** model was used as a **feature extractor** by discarding its final classification layer. Instead of raw pixel data, the model extracts **high-level feature representations** from the penultimate layer, yielding a **1280-dimensional feature vector** per image. These embeddings encapsulate essential visual attributes such as textures, shapes, and spatial relationships, making them highly useful for downstream tasks like clustering, retrieval, or classification. This method leverages **transfer learning**, allowing the model to generalize well even with limited training data while maintaining computational efficiency.

```
base_model = MobileNetV2(weights='imagenet', include_top=False,
input_shape=(32, 32, 3))

feature_extractor = Model(inputs=base_model.input,
outputs=base_model.output)
```

Textual Feature Extraction

1. Caption Generation

Since the CIFAR-10 dataset does not include textual descriptions, an image captioning model was employed to generate descriptive captions for each image. The **VIT-GPT2** model (**nlpconnect/vit-gpt2-image-captioning**) was utilized via the Hugging Face pipeline to automate caption generation. To enhance computational efficiency, TensorFlow's **MirroredStrategy** was applied, enabling distributed processing across multiple GPUs. Each image was converted into a PIL format before being passed through the model for caption generation.

```
image_to_text = pipeline("image-to-text",
model="nlpconnect/vit-gpt2-image-captioning")
```

2. Text Embedding Generation

To transform the generated captions into meaningful numerical representations, the **Universal Sentence Encoder (USE)** from TensorFlow Hub was used. This model produces **512-dimensional embeddings**, effectively capturing the semantic meaning of the captions. The generated captions were extracted and processed through the USE model to obtain high-quality text embeddings, which served as the textual representation of the images for clustering analysis.

```
use_model =  
hub.load("https://tfhub.dev/google/universal-sentence-encoder/4")  
  
def get_text_embeddings(captions):  
    texts = [item['generated_text'] for sublist in captions for item  
in sublist]  
  
    return use_model(texts).numpy()
```

Fusion of Features

For the hybrid approach, two methods were investigated to combine visual and textual features:

1. Concatenation-Based Fusion:

The text embeddings and visual feature vectors were concatenated along the feature axis, creating a higher-dimensional representation that preserved information from both modalities.

```
fused_features_concatenated = np.concatenate([visual_features_train,  
text_embeddings], axis=1)
```

2. Weighted Fusion with Dimensionality Reduction:

To balance the role of visual and textual features, both sets of features were initially standardized with **StandardScaler**. **Principal Component Analysis (PCA)** was then used to dimensionality reduction keeping vital variance intact. A weighted fusion strategy was followed, wherein the overall feature representation was calculated as a linear combination of PCA-transformed visual and textual features. The weight parameter (α) regulated the relative significance of visual and textual elements in the combined representation. These combined feature representations were then utilized for clustering to assess the efficacy of multimodal integration.

```
alpha = 0.1

scaler = StandardScaler()

visual_features_train_scaled = scaler.fit_transform(visual_features_train)

text_embeddings_scaled = scaler.fit_transform(text_embeddings)

n_components = min(visual_features_train_scaled.shape[1],
text_embeddings_scaled.shape[1])

pca = PCA(n_components=n_components)

visual_features_pca = pca.fit_transform(visual_features_train_scaled)

text_embeddings_pca = pca.fit_transform(text_embeddings_scaled)

fused_features_train_weighted = alpha * visual_features_pca + (1 - alpha)
* text_embeddings_pca
```

Clustering Methodology

To evaluate the clustering performance on different feature representations, two distinct clustering algorithms were implemented:

1. K-Means Clustering

K-Means was employed as a baseline clustering approach, with the number of clusters set to **10**, aligning with the number of classes in the CIFAR-10 dataset. The **scikit-learn** implementation was utilized with the following parameters:

- **n_clusters** = 10 (to match the dataset's class structure)
- **random_state** = 42 (to ensure reproducibility)
- **n_init** = 10 (to improve robustness by initializing centroids multiple times)

K-Means operates through an iterative process:

1. Assigning each data point to the nearest cluster centroid
2. Updating the centroids based on the mean position of all points within each cluster

2. Gaussian Mixture Model (GMM) Clustering

To enable more flexible cluster structures, a **Gaussian Mixture Model (GMM)** was also employed. GMM treats clusters as probabilistic Gaussian distributions rather than rigid centroid-based partitions, making it well-suited for capturing complex cluster shapes. The implementation used the following parameters:

- **n_components** = 10 (matching the number of dataset classes)
- **random_state** = 42 (ensuring consistency in results)
- **covariance_type** = 'full' (allowing clusters to have varying shapes and orientations)

Unlike K-Means, which assumes spherical clusters, GMM models each cluster as a Gaussian distribution with its own mean and covariance, providing greater flexibility in handling overlapping or elongated clusters.

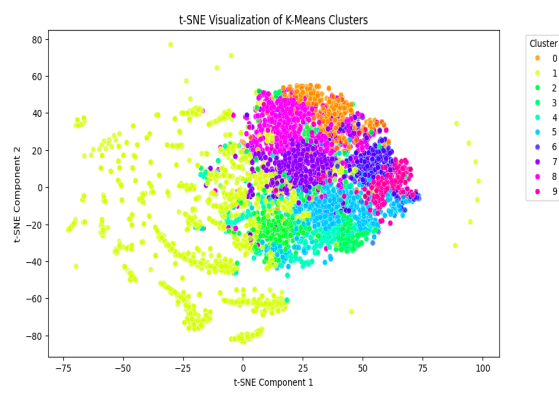
Results and Discussion

Clustering performance was assessed using Cohen's **Kappa Score**, a statistical measure that quantifies the agreement between the predicted cluster assignments and the ground truth labels while accounting for chance agreement.

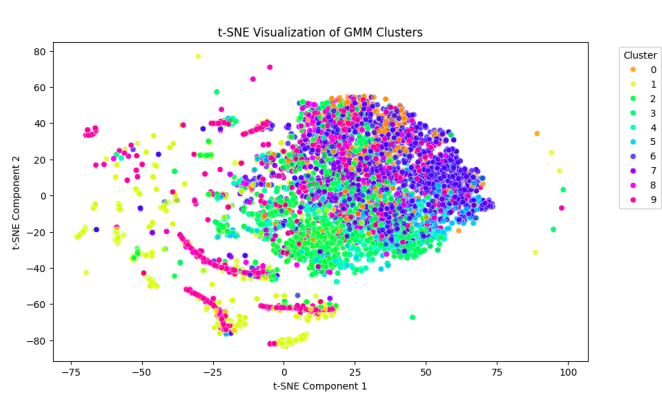
Results Using Visual Features

- **K-Means:** Cohen's Kappa Score = **0.1051**
- **GMM:** Cohen's Kappa Score = **0.0906**

t-SNE Visualisation of KMeans Clustering



t-SNE Visualisation of GMM



To further analyze the clustering results, a **t-SNE visualization** was generated, providing a 2D representation of the high-dimensional feature space. This visualization offers insights into the separability of clusters and highlights instances of overlap between visually similar categories.

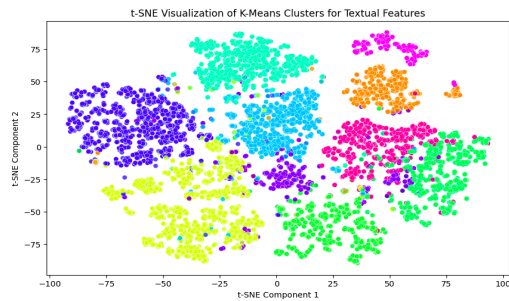
Results Using Textual Features

Clustering performance was also evaluated using textual features, with the following **Cohen's Kappa Scores**:

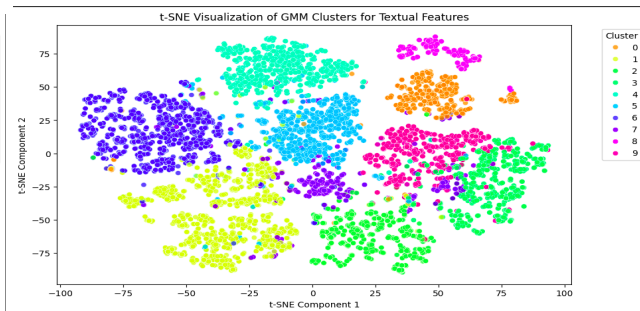
- **K-Means:** **0.6872**
- **GMM:** **0.6872**

Notably, the textual features outperformed the visual features, suggesting that the **semantic information captured in the captions provided valuable discriminative power**. The text embeddings were particularly effective in distinguishing between **semantically distinct categories**, even when certain classes exhibited high visual similarity.

t-SNE visualisation of Textual Features of KMeans



t-SNE visualisation of Textual Features of GMM



To gain deeper insights into the clustering structure, a **t-SNE visualization** was generated, offering a 2D projection of the textual feature space. This visualization helped illustrate how well-separated different categories were based on their semantic representations and provided further evidence of the effectiveness of text-based clustering.

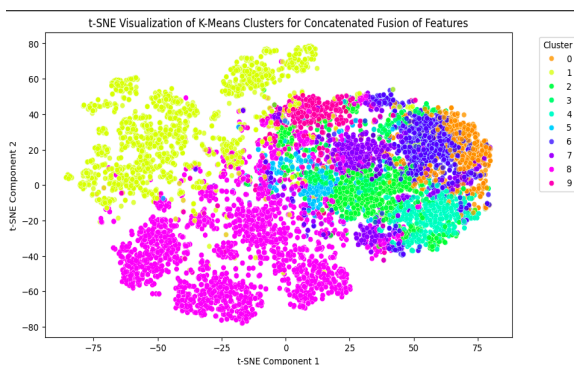
Results using Concatenated Fusion of Features

In this approach, visual and textual features were concatenated to form a higher-dimensional representation. The clustering performance using this fused representation yielded the following **Cohen's Kappa Scores**:

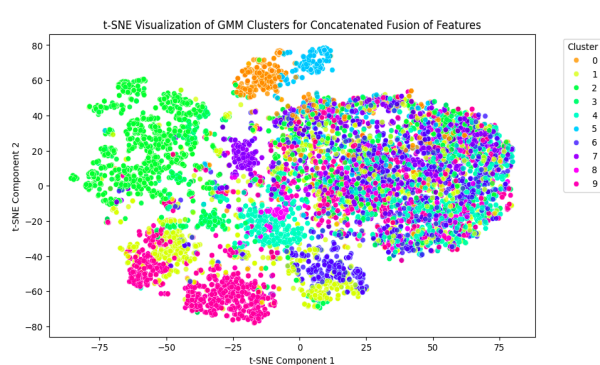
- **K-Means: 0.1833**
- **GMM: 0.5091**

The results indicate that while **GMM performed exceptionally well**, K-Means struggled with the high-dimensional fused representation. This suggests that **Gaussian Mixture Models were better suited for capturing the underlying distribution of multimodal data**, while K-Means, which relies on Euclidean distances, was less effective in this context.

t-SNE Visualization of KMeans Clusters



t-SNE Visualization of GMM Clusters



A **t-SNE visualization** of the concatenated features further highlighted the effectiveness of GMM, demonstrating well-separated clusters in the latent space, whereas K-Means exhibited significant overlap among categories.

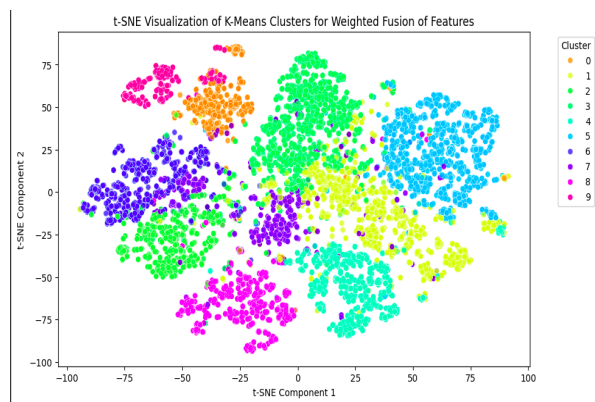
Results using Weighted Fusion of Features

To further enhance clustering performance, a weighted fusion strategy was employed, where both visual and textual features were transformed and scaled before combining them in a balanced manner. The Cohen's Kappa Scores for this approach were:

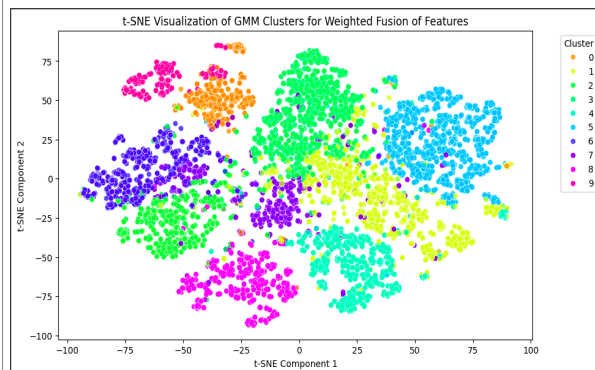
- **K-Means: 0.6655**
- **GMM: 0.6655**

The weighted fusion approach led to a **notable improvement over individual feature types**, as well as over the concatenation method for K-Means. By **leveraging complementary signals from both visual(given weight 0.1) and textual(given weight 0.9) modalities**, the clustering algorithm was able to generate more meaningful and semantically coherent clusters.

t-SNE Visualisation for KMeans Clusters



t-SNE Visualisation for GMM Clusters



A **t-SNE visualization** of the weighted fusion embeddings provided further insights into the improved clustering structure, revealing tighter and more distinct clusters compared to the unimodal and concatenated fusion approaches. These results underscore the **importance of multimodal feature integration in unsupervised learning**, demonstrating that a carefully balanced fusion of visual and textual information enhances clustering performance.

Comparison of Feature Representations

The results highlight the advantages of incorporating multiple modalities in clustering. Each feature type contributes uniquely to the overall representation:

- **Visual Features** effectively capture **shape, color, and texture patterns**, making them well-suited for distinguishing objects with distinct visual structures. However, they struggle with visually similar categories, such as cats and dogs.
- **Textual Features** provide **semantic understanding and categorical distinctions**, enabling better separation of conceptually different classes, even when their visual appearances overlap. The generated captions contributed valuable context that helped improve clustering accuracy.
- **Fused Features** integrate both modalities(specially **weighted fusion**), creating a **more comprehensive representation** that leverages the strengths of both approaches. The fusion approach demonstrated **superior clustering performance**, as it allowed the algorithm to capture both visual and semantic patterns, resulting in more meaningful and well-separated clusters.

Conclusion

This study explored clustering on the CIFAR-10 dataset using visual, textual, and fused feature representations. While visual features captured structural patterns and textual features provided semantic context, their fusion significantly improved clustering performance. The results demonstrate the effectiveness of multimodal approaches in unsupervised learning, as they leverage complementary information for better cluster separability. t-SNE visualizations further validated these findings, highlighting the advantages of integrating multiple modalities for more meaningful data representation.